

vision**2**learn

Coding Practices Level 3

Hints
and
Tips

Unit 1



Contents

Section 1 – Understand software methodologies in project management	4
Section 2 – Understand the capture and documentation of requirements	10
Section 3 - Understand risk management	11
Section 4 - Understand how teamwork contributes to effective delivery of software	12

Hints and Tips Unit 1

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 1: Coding Requirements and Planning

Section 1 – Understand software methodologies in project management

Q1. For the methodology “Agile”, answer the following questions:

a) **Summarise** the key features of agile, include the following key points within your answer:

- the key aims of the methodology
- any dependencies/requirements for use
- roles and responsibilities
- how work is divided between team members
- how the Software Delivery Lifecycle is utilised
- tools and processes
- checkpoints and milestones
- how progress is measured
- any impact to the customer

You must ensure that you include each of the key points in the bullet point lists within your summarisation. This will be a minimum of two sentences for each one.

b) **Consider** the strengths and weaknesses of Agile, include the following in your answer:

- requirements collection
- task breakdown
- task workflow
- change management
- the feedback loop

Think carefully of what the strengths and weaknesses of agile are. When writing your answer, be sure to include each point in the bulleted list. Are these a strength or a weakness in terms of scrum? It's likely that each one has a mixture of both strengths and weaknesses and you should write about this.

- c) Use the case study on page 5 of the course content, or if you prefer you may research your own and answer the following:

Give an **example** of when Agile methodology is most suitable, consider the following in your example:

- project and product types
- requirements and documentation
- compliance and governance
- visibility and scope
- plan and timescales
- staffing and resourcing
- volatility

Review the course content on page 5 and case study for Agile methodology. You may also conduct additional research which you can use to support your answer. The bullet points listed above must be considered and detail provided for each within your description of when Agile methodology is most suitable.

Q2. For the methodology “**Scrum**”, answer the following questions:

a) **Summarise** the key features of Scrum, include the following key points within your answer:

- the key aims of the methodology
- any dependencies/requirements for use
- roles and responsibilities
- how work is divided between team members
- how the Software Delivery Lifecycle is utilised
- tools and processes
- checkpoints and milestones
- how progress is measured
- any impact to the customer

You must ensure that you include each of the key points in the bullet point lists within your summarisation. This will be a minimum of two sentences for each one.

b) Consider the strengths and weaknesses of Scrum, include the following in your answer:

- requirements collection
- task breakdown
- task workflow
- change management
- the feedback loop

Think carefully of what the strengths and weaknesses of Scrum are. When writing your answer, be sure to include each point in the bulleted list. Are these a strength or a weakness in terms of scrum? It's likely that each one has a mixture of both strengths and weaknesses and you should write about this.

c) Use the case study on page 5 of the course content, or if you prefer you may research your own and answer the following:

Give an example of when Scrum methodology is most suitable, consider the following in your example:

- project and product types
- requirements and documentation
- compliance and governance
- visibility and scope
- plan and timescales
- staffing and resourcing
- volatility

Review the course content on page 5 and case study for Scrum methodology. You may also conduct additional research which you can use to support your answer. The bullet points listed above must be considered and detail provided for each within your description of when Scrum methodology is most suitable.

Q3. For the methodology “Kanban”, answer the following questions:

a) Summarise the key features of Kanban, include the following key points within your answer:

- the key aims of the methodology
- any dependencies/requirements for use

- roles and responsibilities
- how work is divided between team members
- how the Software Delivery Lifecycle is utilised
- tools and processes
- checkpoints and milestones
- how progress is measured
- any impact to the customer

You must ensure that you include each of the key points in the bullet point lists within your summarisation. This will be a minimum of two sentences for each one.

b) Consider the strengths and weaknesses of Kanban, include the following in your answer:

- requirements collection
- task breakdown
- task workflow
- change management
- the feedback loop

Think carefully of what the strengths and weaknesses of Kanban are. When writing your answer, be sure to include each point in the bulleted list. Are these a strength or a weakness in terms of scrum? It's likely that each one has a mixture of both strengths and weaknesses and you should write about this.

c) Use the case study on page 5 of the course content, or if you prefer you may research your own and answer the following:

Give an example of when Kanban methodology is most suitable, consider the following in your example:

- project and product types
- requirements and documentation
- compliance and governance
- visibility and scope
- plan and timescales
- staffing and resourcing
- volatility

Review the course content on page 5 and case study for Kanban methodology. You may also

conduct additional research which you can use to support your answer. The bullet points listed above must be considered and detail provided for each within your description of when Kanban methodology is most suitable.

Q4. For the methodology “Waterfall”, answer the following questions:

a) Summarise the key features of Waterfall, include the following key points within your answer [1.1]:

- the key aims of the methodology
- any dependencies/requirements for use
- roles and responsibilities
- how work is divided between team members
- how the Software Delivery Lifecycle is utilised
- tools and processes
- checkpoints and milestones
- how progress is measured
- any impact to the customer

You must ensure that you include each of the key points in the bullet point lists within your summarisation. This will be a minimum of two sentences for each one.

b) Consider the strengths and weaknesses of Waterfall, include the following in your answer [1.3]:

- requirements collection
- task breakdown
- task workflow
- change management
- the feedback loop

Think carefully of what the strengths and weaknesses of waterfall are. When writing your answer, be sure to include each point in the bulleted list. Are these a strength or a weakness in terms of scrum? It's likely that each one has a mixture of both strengths and weaknesses and you should write about this.

c) Use the case study on page 5 of the course content, or if you prefer you may research your own and answer the following:

Give an **example** of when Waterfall methodology is most suitable, consider the following in your example [1.4]:

- project and product types
- requirements and documentation
- compliance and governance
- visibility and scope
- plan and timescales
- staffing and resourcing
- volatility

Review the course content on page 5 and case study for Waterfall methodology. You may also conduct additional research which you can use to support your answer. The bullet points listed above must be considered and detail provided for each within your description of when Waterfall methodology is most suitable.

Q5. Using the information for your answers above, **describe** the similarities and differences of the methodologies. You should make reference to all four methodologies in your answer:

Using all four of the methodologies: scrum, Kanban, waterfall and agile, describe what aspects of each one are similar and what ones are different. You could include when you might use each one and when more than one could be used for the same situation.

Section 2 – Understand the capture and documentation of requirements

Q1. Identify the process of requirements capture:

As an identify question, you may list the elements including with the process of requirements capture with a few words added to demonstrate your understanding of each point.

Q2. Describe the following terms:

- Epics
- Features
- User stories
- Backlogs

Describe each term within the table provided. Don't be tempted to copy and paste definitions here. You should use your own words to ensure a description of each is given. You could use examples of when their used if this helps with your answer.

Q3. On page 2 of the course content you are given a range of alternative documentation methods, select *two* of these and **explain** when would you use them, why and the benefits/drawback of each method.

Review the 'read' document on page 2 of the course content, which contains information on alternative methods of document requirements. Select 2 of these (you may use others that you're familiar with if you prefer) and explain when you would use each one. Consider that the requirements are not always complete or hold all required information.

Section 3 - Understand risk management

Q1. Explain what is meant by risk management:

This is an explain question and will therefore need a detailed answer. You should state what risk management is and then explain the reasons for its use.

Q2. Describe potential risks that could affect project delivery:

This is a describe question and will therefore need a detailed answer. You should at least list three different risks within your answer and fully describe how each one could affect project delivery.

Q3. Describe how to reduce risk and follow mitigation for each of the following areas. Ensure that you refer to the key methodologies (Scrum, Kanban, Waterfall and Agile) within your answer:

- skills shortage
- time
- budget
- resource shortage
- decision-making

Each bullet pointed item above is titled within the table. You must describe how to reduce risk for each one and what mitigation should be implemented. Within each description refer to the key methodologies (Scrum, Kanban, Waterfall and Agile) that are best suited to your solution.

Q4. Explain what is meant by a project's critical path:

This is an explain question and will therefore need a detailed answer. You should state what a project's critical path is and then explain it in detail.

Section 4 - Understand how teamwork contributes to effective delivery of software

Q1. Read the case study above and **explain** why the three teams need to develop good working relationships with each other and what the benefits of this are:

You are provided with a case study in the assessment for the company Code4U. You should explain why the teams in the case study need to work together closely and develop a good relationship with each other. It's important to also include the benefits to the teams, the project and the organisation as a whole for everyone working well together. You may consider other teams/departments that could also be included.

Q2. Read the case study above and **explain** the following:

a) The conflict management techniques that may be used for effective team collaboration:

The original case study continues in this question of the assessment and you should use this to consider what management techniques could be employed to assist the teams to overcome the difficulties they are experiencing. As this is an explain question, you should state your ideas and expand on why you think they should be used and what affect you expect them to have on the situation.

b) The potential impact of ineffective working relationships with other teams and departments:

This is another explain question based upon the continued case study. Consider the current status of the teams and explain the difficulties that they are having with communication and working towards the project objectives. Remember that as an explain question, you should give reasons for any points that you make.

c) How team members can reflect on progress and identify improvements:

A further question for the continued case study that also requires an explanation. For this one you should assume that some of the techniques that you suggested in 2a) have been effective and the team is beginning to work well again. To progress with this development, they should review what worked well and what should still be implemented to allow them to work harmoniously in all future projects. Remember that as an explain question, you should give reasons for any points that you make.